AML KNOWLEDGE LIBRARY

Red Flag Indicators & Typology Database

Documentation & User Manual | Prototype v1.0

Total Indicators 144	**High Risk** 114	**OCG Typologies** 10
Indicator Categories 19	**Medium Risk** 30	**Regulatory Sources** 7

1. Overview

The AML Knowledge Library is a structured, queryable repository of anti-money laundering (AML) red flag indicators and typologies. It is designed to serve as a dual-purpose resource supporting both quantitative model development and broadly, qualitative investigative analysis.

The library aggregates, standardizes, and annotates indicators drawn from regulatory guidance and financial intelligence unit (FIU) operational alerts. All indicators are traceable to their primary regulatory source and enriched with metadata fields that support detection logic design, peer comparison, and typology mapping.

1.1 Scope & Emphasis

The library emphasizes Organized Crime Group (OCG)-related typologies, with a focus on the following criminal networks and schemes:

- Chinese Money Laundering Networks (CMLN): mule account patterns and TBML schemes
- Human Trafficking for Sexual Exploitation (HT-SEX): financial patterns of trafficking operators
- Mexican Cartel Oil Smuggling (OIL): revenue laundering through O&G and freight sectors
- Synthetic Opioid Networks (FENT): opioid production, procurement, and distribution financing
- Professional Money Laundering (PML): TBML and MSB exploitation by specialist ML networks
- Underground Banking (UB): informal value transfer including hawala and alternative remittance

To ensure a feasible scope, only indicators applicable to the specific datasets provided were collected within the typologies. The library however also includes a foundation of general ML/TF indicators (drawn from FINTRAC's General Indicators guidance) that underpin all typologies and support broad-spectrum model development; these indicators are non-specific to the data.

1.2 Library Composition

Category	Count	Description
Profile Mismatch	18	Client financial profile inconsistencies relative to declared occupation, income, or business
Underground Banking	15	ATH-sector and hawala scheme indicators
Layering	14	Transaction structuring to create distance between funds and illicit origin
Structuring	9	Deliberate transaction fragmentation to avoid reporting thresholds
Wire Transfers	10	Suspicious international and domestic wire transfer patterns
Account Activity	10	Anomalous account behavior changes over time
Atypical Transactions	9	Transactions lacking economic rationale or matching criminal typologies
Crimes & High-Risk Entities	14	HT-SEX, illicit storefronts, and victim maintenance financial patterns
Geographic Risk	8	High-risk jurisdiction and corridor exposure
Production & Procurement	6	Illicit drug production and supply procurement financing
Trade-Based ML	7	TBML through commodity manipulation and invoice fraud
Client Identification	6	Identity fraud and document inconsistency indicators

Terrorist Financing	3	Conflict zone transactions and travel financing
Bribery & Corruption	4	PEP-related, offshore commission, and industry bribery indicators
Distribution & Corridors	5	Drug distribution network financing and corridor-based trafficking
Politically Exposed Persons	2	PEP-specific wealth and asset concealment indicators
Products & Services	1	Anomalous use of specific financial products
Gatekeeper Misuse	1	Professional account misuse by lawyers, accountants, and notaries
Client Behavior	1	Unusual in-branch or transactional behaviors
Placement	1	Multi-location cash placement indicators
TOTAL	144	

2. User Manual

Note that the fields **Quantifiable_Threshold, Data_Relevance,** and **Detection_Difficulty** have been removed from the final version of the knowledge library due to specificity and complexity; users must add and determine these fields independently as they depend primarily on the data provided.

2.1 Modelling Experts

Primary use case: Building, calibrating, and validating transaction monitoring rules, machine learning features, and AML detection models.

Key Fields for Model Development

Field	Description / Values
Quantifiable_Threshold	Machine-readable condition or formula (e.g., "Amount MOD 50 = 0", "Credits > 200% of income") that can be directly translated into rule logic or used as a model feature definition.
Model_Applicability	The detection method(s) best suited to identifying the indicator (e.g., Transaction Monitoring, Velocity Rules, Network Analysis, Peer Comparison).
Data_Relevance	Describes which data assets or fields enable detection — directly maps to data pipelines (e.g., "ABM city/province and timestamps available").
Detection_Difficulty	Easy / Moderate / Hard classification indicating how challenging automated detection is without additional context.
Co_Occurring_Indicators	IDs of other indicators that tend to co-occur, enabling multi-signal rule design and compound scoring.
Risk_Level	High / Medium / Low — informs alert scoring and prioritization logic.
Transaction_Type	Narrows which transaction data source (ABM, Card, Wire, EMT, EFT, Cheque, etc.) the indicator applies to.

Sample Workflow

1. Filter by **Model_Applicability** to identify indicators matching the detection approach (e.g., all Velocity Rules indicators).
2. Use **Quantifiable_Threshold** as a starting point for rule definition; calibrate thresholds against the given data distribution.
3. Use **Co_Occurring_Indicators** to design compound rules or ensemble features that require multiple signals to trigger.
4. Reference **Data_Relevance** to confirm that required data assets are available in the pipeline before implementing an indicator.
5. Use **Source_URL** to access original guidance for full regulatory context when designing model documentation.

2.2 AML Investigators

Primary use case: Alert investigation, SAR/STR drafting, typology research, and escalation triage.

Key Fields for Investigation

Field	Description / Values
Red_Flag_Description	Plain-language description of the indicator as it would appear in a case.
Notes	Contextual intelligence note explaining the typological significance of the indicator and common co-occurring criminal behaviors.
OCG_Association	The criminal organization type(s) most commonly associated with this indicator — useful for typology matching in STR narratives.
Geographic_Focus	The geographic scope where this indicator is most prevalent or where regulatory guidance specifically applies.
Co_Occurring_Indicators	IDs of related indicators to cross-reference when building a case or identifying pattern clusters.
Source_Primary	The regulatory or intelligence source, providing authority for citing the indicator in STR narratives or escalation memos.

Sample Workflow

6. Start with **By_Risk** tab (Excel) or filter by Risk_Level = High to prioritize indicators for triage.
7. Use **By_Category** or **Typology_Groups** to navigate to the relevant criminal scheme when investigating a specific case type.
8. Reference **Notes** field for narrative language and typological context suitable for use in STR reports.
9. Use **Co_Occurring_Indicators** to identify related indicators and build a multi-signal case narrative.
10. Cite **Source_Primary** and **Source_URL** in STR narratives or escalation memos to establish regulatory authority.

2.3 Other User Groups

Compliance Officers & Policy Teams

- Use the Sources sheet to identify which regulatory advisories have been incorporated and verify currency of coverage.
- Cross-reference indicator categories against internal policy documents to identify coverage gaps.
- Use **Risk_Level** and **Detection_Difficulty** fields to prioritize policy updates and staff training.

Intelligence & Research Analysts

- Use **Typology_Groups** (Excel) or Section 5 of this document to understand OCG-specific financial patterns.
- Use **OCG_Association** and **Geographic_Focus** to map intelligence to geographic or organizational targets.
- Use **Co_Occurring_Indicators** to build typology clustering maps for strategic intelligence products.

Training & Awareness

- **Red_Flag_Description** and **Notes** fields provide plain-language, training-ready indicator descriptions.
- Indicators can be filtered **by Indicator_Category** to build category-specific training modules.
- **By_Category** sheet in the Excel workbook provides a structured, visually organized format suitable for presentation.

3. Methodology

3.1 Indicator Extraction

Indicators were extracted from primary regulatory guidance documents and FIU intelligence publications. Each source document was reviewed in full, with red flag indicators manually identified, isolated, and mapped to a standardized schema. Where source documents provided narrative descriptions without discrete indicator demarcation, indicators were decomposed into individual, independently actionable signals. Many indicators were incompatible with the data provided; if too few indicators in a category were applicable, all would be removed.

3.2 Standardization

All extracted indicators were standardized across the following dimensions:

- **Category and Sub-Category**: Indicators were assigned to one of 19 standardized typology categories. Where source documents used inconsistent or overlapping categories, indicators were reclassified based on the primary nature of the signal (e.g., pass-through patterns were consolidated under Layering regardless of original source categorization).
- **Risk Level**: Risk ratings were assigned based on the regulatory emphasis, typological specificity, and detection confidence of each indicator. Notably, the *Low* rating was ultimately removed entirely.
- **Identifier Scheme**: All indicators received a unique alphanumeric ID following the pattern [PREFIX]-[SEQUENCE] (e.g., STRUCT-001, HT-SEX-07). Prefix codes reflect the typology group or source context.
- **Date Standardization:** All publication dates normalized to ISO 8601 format (YYYY-MM-DD).
- **Separator Standardization:** All Co_Occurring_Indicators fields use comma-space separated lists.

3.3 Enrichment

Each indicator was enriched with fields beyond those provided in the source documents:

- **Quantifiable_Threshold:** A machine-readable operationalization of the indicator, expressed as a condition, formula, or comparison (e.g., "Credits > 200% of Stated Income"). These thresholds are starting points and require institutional calibration.
- **Model_Applicability:** Annotated based on the detection methodology implied by the indicator's nature (e.g., velocity-based indicators → Velocity Rules; multi-account patterns → Network Analysis).
- **Data_Relevance:** Added to indicate which data assets enable detection, directly supporting data pipeline scoping.
- **Notes:** Contextual intelligence annotation added for all indicators, drawing on typological knowledge and regulatory guidance to provide investigative and model-development context.

3.4 Quality Controls

- While indicators in the specific OCG-related typologies were individually chosen given applicability to the specific data provided, indicators in the categories ID-001 – TF-003 were deliberately preserved in their entirety from the original documentation to preserve broader AML context.
  - It should hence be noted that indicators from CMLN-MULE-01 onwards are case-specific and unsuitable for repurposing.
  - Although distinct from money laundering, Terrorist Financing was also included as an indicator category for similar reasons, as it is investigated by FINTRAC in conjunction with money laundering.
- There are intentional redundancies in indicators between the general and OCG-specific categories; the distinction is preserved to better isolate OCG typologies where an AI model may not set appropriate boundaries between OCG associations.
- Duplicate sub-category resolutions were made; for example, STRUCT-007 and STRUCT-009 originally shared the same sub-category "Threshold Knowledge" when identified. These were differentiated into "Threshold Inquiry" (client actively asking about thresholds) and "Threshold Awareness" (client demonstrating awareness through behavior) to preserve distinct investigative significance.

4. Data Sources

All indicators in the library are traceable to one of the following seven primary regulatory sources. No secondary or unverified sources were used.

1. FINTRAC - General ML/TF Indicators

Issuing Organization	FINTRAC (Financial Transactions and Reports Analysis Centre of Canada)
Document Title	General ML/TF Indicators — Guidance for Reporting Entities
Publication Date	2025-11-24
Jurisdiction	Canada
Indicators Indexed	63
URL	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng
Summary	Comprehensive FINTRAC guidance listing general money laundering and terrorist financing red flag indicators applicable to all reporting entities under the Proceeds of Crime (Money Laundering) and Terrorist Financing Act (PCMLTFA). Covers client identification, behavior, profile mismatch, structuring, wire transfers, geographic risk, bribery, PEPs, and TF.

2. FINTRAC 2021-OA-HTS

Issuing Organization	FINTRAC
Document Title	Operational Alert: Human Trafficking for Sexual Exploitation
Publication Date	2021-07-01
Jurisdiction	Canada
Indicators Indexed	14
URL	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-eng
Summary	Operational alert identifying the financial indicators most commonly associated with human trafficking for sexual exploitation in Canada. Covers retail and gift card spend, victim logistics, digital integration, geographic travel patterns, illicit storefront operations, and asset procurement typologies.

3. FinCEN FIN-2025-A003

Issuing Organization	FinCEN (Financial Crimes Enforcement Network, U.S. Treasury)
Document Title	Advisory on Chinese Money Laundering Networks
Publication Date	2025-08-28
Jurisdiction	United States
Indicators Indexed	9
URL	https://www.fincen.gov/resources/advisories/fincen-advisory-fin-2025-a003

Summary	FinCEN advisory identifying red flags associated with Chinese money laundering networks (CMLN) exploiting U.S. financial institutions. Covers money mule patterns (including student visa holders), TBML through e-commerce fronts, Daigou networks, credit card clearing, and front business characteristics.

4. FinCEN FIN-2025-Alert002

Issuing Organization	FinCEN (Financial Crimes Enforcement Network, U.S. Treasury)
Document Title	Alert: Oil Smuggling on the U.S. Southwest Border
Publication Date	2025-05-01
Jurisdiction	United States
Indicators Indexed	7
URL	https://www.fincen.gov/system/files/shared/FinCEN-Alert-Oil-Smuggling-FINAL-508C.pdf
Summary	FinCEN alert on cartel oil smuggling revenue laundering through U.S. oil and gas, freight, and import/export sectors. Identifies CJNG and Sinaloa cartel financial patterns including sector outlier activity, waste oil invoicing fraud, fund mirroring, shell entity characteristics, and residential business registration.

5. FINTRAC 2025-OA-001

Issuing Organization	FINTRAC
Document Title	Operational Alert: Synthetic Opioid Trafficking
Publication Date	2025-01-01
Jurisdiction	Canada
Indicators Indexed	18
URL	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
Summary	FINTRAC operational alert on the financial patterns associated with synthetic opioid (fentanyl) trafficking networks. Covers illicit lab production procurement, clandestine logistics, geographic distribution corridors, and digital layering techniques including EMT pass-through, gambling masking, and structured cash deposits.

6. FINTRAC 18/19-SIDEL-025

Issuing Organization	FINTRAC
Document Title	Strategic Intelligence: Professional Money Laundering — TBML & MSB Networks
Publication Date	2018-07-18
Jurisdiction	Canada
Indicators Indexed	18

URL	https://fintrac-canafe.canada.ca/intel/operation/oai-ml-eng
Summary	FINTRAC strategic intelligence report on professional money laundering (PML) networks using trade-based ML and MSB exploitation. Covers commodity risk, invoice manipulation, geographic mirroring, MSB pass-through, personal misuse of MSB accounts, and PML-MSB nexus patterns.

7. FINTRAC 2023-OA002

Issuing Organization	FINTRAC
Document Title	Operational Alert: Underground Banking — ATH Schemes
Publication Date	2023-07-01
Jurisdiction	Canada
Indicators Indexed	15
URL	https://fintrac-canafe.canada.ca/intel/operation/ml-rec-eng
Summary	FINTRAC operational alert on underground banking schemes using the Alternative Transfer and Hawala (ATH) sector. Covers sector velocity indicators, fan-out distribution, pass-through accounts, MSB nexus, signatory funding, and asset conversion patterns through real estate, securities, automotive, and legal sectors.

5. Design Decisions, Dependencies & Assumptions

5.1 Data Model Design

The library uses a flat, row-per-indicator schema with 17 fields. This design was chosen to maximize compatibility with the widest range of tools (CSV, Excel, SQL, Python/pandas, JSON) without requiring relational joins. Co-occurrence relationships are represented as a comma-separated list of Indicator IDs within each row rather than as a separate relationship table, accepting minor redundancy in exchange for usability.

5.2 Category Architecture

Indicator categories represent the primary typological nature of the signal, not the criminal offense or the source document's original classification. For example, all pass-through patterns are classified under Layering regardless of whether they originated from HT, OIL, or FENT guidance. This cross-typology normalization enables model developers to apply layering-specific detection logic to all relevant indicators simultaneously.

5.3 Quantifiable Thresholds

Assumption: Thresholds are illustrative starting points derived from regulatory guidance language, not validated statistical thresholds. They require calibration against institutional transaction data before use in production models. Pseudo-code syntax (e.g., "Amount MOD 50 = 0", "Credits > 200% of income") is used for clarity; translation to specific query languages or ML feature engineering code is the implementer's responsibility.

5.4 Risk Level Assignment

Risk levels reflect the typological significance and actionability of the indicator in the context of a consolidated, OCG-focused library. Adjustments were made from source-implied levels case-by case. Sample changes:

- PROF-009 (Out-of-Area Account Opening): Low → Medium. A Low risk designation is unsuitable in a library where all indicators represent genuine red flags; out-of-area account opening is a meaningful signal requiring contextual assessment.
- WIRE-001 (Unknown Wire Counterparty): Medium → High. Not knowing the sender or recipient of a wire transfer is a primary ML indicator, particularly in a mule account context.
- CORRUPT-004 (Opaque New Entities): Medium → High. Transacting with newly established or opaque entities is a significant indicator of shell company facilitation, warranting High classification.

5.5 Dependencies

- Excel Workbook (.xlsx): Requires Microsoft Excel 2016+ or equivalent (LibreOffice Calc, Google Sheets) for full filter and formatting functionality. Conditional formatting applied using openpyxl.
- CSV file: Encoded in UTF-8. Compatible with all standard data tools. Delimiter is comma; text fields containing commas are double-quoted.
- Source URLs: All URLs were valid as of the library's compilation date. Regulatory websites may restructure URLs; use the **Source_Primary** reference name to locate documents if links break.
- Data Currency: Indicators reflect guidance current as of the publication dates listed. New regulatory advisories or typology updates should be incorporated by adding new indicator rows following the existing ID naming convention.

5.6 Out of Scope

- Sanctions screening indicators are not included; these are typically addressed through separate list-matching systems.
- Indicators specific to virtual asset service providers (VASPs) or cryptocurrency-native typologies are not systematically covered; some digital indicators appear incidentally within broader typologies.
- Fraud-specific indicators (e.g., account takeover, social engineering) are excluded except where directly co-occurring with AML typologies.

6. Glossary

Definitions for all abbreviations, shorthand, and technical terms used in this library.

6.1 Abbreviations & Acronyms

Term	Definition	Notes / Context
ABM	Automated Banking Machine (Canadian equivalent of ATM)	*Used in transaction type and geographic analysis fields*
AML	Anti-Money Laundering	*Regulatory framework and compliance discipline*
ATH	Alternative Transfer and Hawala	*Underground banking sector designation used by FINTRAC*
CAD	Canadian Dollar	*Currency reference for FINTRAC reporting thresholds ($10K CAD)*
CJNG	Cartel Jalisco Nueva Generación	*Major Mexican drug trafficking organization*
CMLN	Chinese Money Laundering Networks	*Typology designation per FinCEN FIN-2025-A003*
CTR	Currency Transaction Report	*U.S. FinCEN mandatory report for cash transactions > $10K USD*
EFT	Electronic Funds Transfer	*Interbank digital transfer method*
EMT	Electronic Money Transfer (e-transfer)	*Peer-to-peer digital payment; e.g., Interac e-Transfer in Canada*
FATF	Financial Action Task Force	*International AML/CTF standard-setting body; maintains grey/blacklists*
FENT	Fentanyl / Synthetic Opioid	*Typology prefix for synthetic opioid-related indicators*
FinCEN	Financial Crimes Enforcement Network	*U.S. Treasury bureau responsible for AML/CFT regulation*
FINTRAC	Financial Transactions and Reports Analysis Centre of Canada	*Canadian Financial Intelligence Unit (FIU)*
FIU	Financial Intelligence Unit	*Government body collecting and analyzing financial intelligence*
FX	Foreign Exchange	*Currency conversion transactions*
HT	Human Trafficking	*Typology prefix for human trafficking indicators*
KYC	Know Your Customer	*Customer due diligence process for identity and profile verification*
MCC	Merchant Category Code	*Four-digit code classifying the type of goods/services a merchant sells*
ML	Money Laundering	*Concealing the origins of illegally obtained money*
MSB	Money Service Business	*Licensed entity offering currency exchange, remittance, or payment services*

OCG	Organized Crime Group	*Structured criminal network engaged in sustained illegal activity*
O&G	Oil and Gas	*Industry sector reference used in OIL-prefix indicators*
P2P	Peer-to-Peer	*Direct digital transfers between individuals*
PCMLTFA	Proceeds of Crime (Money Laundering) and Terrorist Financing Act	*Canadian primary AML legislation*
PEP	Politically Exposed Person	*Individual holding prominent public function; elevated ML/corruption risk*
PML	Professional Money Launderer	*Specialist criminal providing ML services to third-party OCGs*
POS	Point of Sale	*In-person card transaction at a merchant terminal*
RE	Real Estate	*Property sector; frequently used for ML integration*
SAR	Suspicious Activity Report	*U.S. FinCEN mandatory report for suspicious financial activity*
SMB	Small and Medium-Sized Business	*Business size classification used in peer benchmarking*
STR	Suspicious Transaction Report	*Canadian FINTRAC equivalent of a SAR*
TBML	Trade-Based Money Laundering	*Laundering proceeds through manipulation of international trade*
TF	Terrorist Financing	*Financing of terrorist individuals, groups, or activities*
UB	Underground Banking	*Informal value transfer operating outside regulated financial system*
USD	United States Dollar	*Currency reference; relevant for FinCEN-sourced thresholds*
WU	Western Union	*Major international money transfer operator; cited in mixed-channel typologies*

6.2 Technical Terms & Concepts

Term	Definition
Beneficial Owner	The natural person(s) who ultimately owns or controls a legal entity or arrangement; a key KYC/CDD concept.
Co-Occurring Indicators	Other indicator IDs that frequently appear alongside a given indicator in ML schemes; used for compound rule design.
Daigou	Grey-market purchasing agents who buy goods in one country for resale in another; exploited in CMLN schemes.
Detection Difficulty	Easy / Moderate / Hard classification of how challenging automated detection is without additional context.

Dormant Account	An account with no transactional activity for 6+ months; frequently reactivated by OCGs for mule use.
Fan-In / Fan-Out	Consolidation of funds from many sources (fan-in) or dispersal to many recipients (fan-out); common underground banking pattern.
Front Company	A legitimate-appearing business used primarily to commingle and launder criminal proceeds.
Gatekeeper	A professional (lawyer, accountant, notary) whose services can be misused to facilitate ML; a FATF-defined risk category.
Integration	The third and final phase of ML: illicit funds re-enter the legitimate economy.
Layering	The second phase of ML: creating distance between funds and their illicit origin through complex transactions.
Mirror Trade	Simultaneous buy/sell transactions in different currencies or jurisdictions used to transfer value without physical movement.
Money Mule	An individual who transfers or receives illicit funds on behalf of criminals, knowingly or unknowingly.
Nominee	A person acting as front owner of assets or accounts on behalf of the true beneficial owner.
Pass-Through	An account used as an intermediary conduit; funds flow in and out with minimal or no residual balance.
Placement	The first phase of ML: introducing illicit cash into the financial system.
Red Flag Indicator	An observable financial behavior or pattern associated with elevated ML/TF risk.
Shell Company	A legal entity with no genuine business operations, used to obscure ownership and transfer value.
Smurfing	A form of structuring where multiple individuals each make small deposits below reporting thresholds.
Structuring	Deliberately breaking transactions into smaller amounts to avoid currency reporting thresholds; a criminal offence.
Typology	A documented pattern or method used by criminals to commit ML/TF; forms the basis for indicator design.
Underground Banking	Informal value transfer networks operating outside regulated financial infrastructure (e.g., hawala, ATH sector).